

well as further theory are referred to Agresti (2002), De Groot (1986), Green (2000), and Mittelhammer, Judge and Miller (2000). Additionally, Journal of Trading's "Statistical Methods to Compare Algorithmic Performance" (2007) gives additional background and examples for the Mann-Whitney U test and the Wilcoxon signed rank test. We follow the mathematical approach presented in the JOT article below.

Each of these approaches consists of: (1) devising a hypothesis, (2) the calculation process to compute the test statistic, and (3) comparing that test statistic to a critical value.

Paired Samples

For paired samples the analysis will split the order into two equal pieces and trade each in a different algorithm over the same exact time horizon. It is important in these tests to only use algorithms that do not compete with one another such as VWAP, TWAP, or POV. A static trade schedule algorithm could also be used in these tests since the strategy is predefined and will not compete with another. The comparison metric used in these tests can be either arrival cost or VWAP performance.

Sign Test

The sign test is used to test the difference in sample medians. If there is a statistical difference between medians of the two paired samples we conclude that the algorithms are not equivalent.

Hypothesis:

H_0 : Medians are the same ($p = 0.5$)

H_1 : Medians are different ($p \neq 0.5$)

Calculation Process:

1. Record all paired observations.
 (X_i, Y_i) = paired performance observations for algorithms X and Y.
 Let $Z_i = X_i - Y_i$.
 k = number of times $Z_i > 0$.
 n = total number of pairs of observations.
2. T is the probability that $z \geq k$ using the binomial distribution

$$T = \sum_{j=k}^n \binom{n}{j} \cdot p^j \cdot (1-p)^{n-j} = \sum_{j=k}^n \binom{n}{j} \cdot (0.5)^j \cdot (0.5)^{n-j}$$

For a large sample the normal distribution can be used in place of the binomial distribution.